

AEGIS v3.1

The Autonomous, Risk-Governed Options Income Desk

Five AI agents run the complete Wheel-strategy lifecycle on Alpaca — discover, sell premium, manage, hedge, and close — with live news grounding, a real VIX-proxy stress index, and a sovereign Risk Agent that actually pulls the veto trigger.

5

AI AGENTS

6

LLM PROVIDERS

11

MCP TOOLS

10

CLI COMMANDS

16

TESTS PASSING

Team ApexArbitrage · Zunurain Tahir · Abdullah Khalid · Loraine Pardillo

Why Most AI Trading Bots Lose Money

Typical hackathon submissions share three structural failure modes:

01

Directional Bets, No Hedge

A single LLM reads RSI + news, buys calls, and prays. One adverse macro headline erases a week of gains. High variance is not a strategy.

02

Opens Trades, Never Manages Them

The bot sells premium and walks away. No take-profit, no stop-loss, no roll, no expiry handling — paper P&L never becomes realized P&L.

03

Disconnected From Reality

Sentiment from stale model memory, hardcoded 'stress index' values, fake order confirmations, and stats typed by hand instead of measured.

AEGIS answer: governed premium harvesting + full lifecycle management + everything measured live.

The Solution — A Complete Autonomous Trading Organization

AEGIS is a five-agent AI trading desk that runs the Wheel strategy end-to-end on Alpaca: sell cash-secured puts (0.15–0.30 Δ) and covered calls (0.20–0.35 Δ) on SPY, QQQ, AAPL, MSFT, NVDA, AMZN — then manage every position to a rule-based exit, guarded by a sovereign Risk Agent.



Live News Grounding

Google News RSS ingestion feeds every LLM call — regime classification, per-ticker sentiment, and earnings blackouts use real-time headlines, never stale training data.



Full Lifecycle Management

Portfolio Manager runs first in every cycle: 50% take-profit, 200% stop-loss, ITM rolls at ≤ 5 DTE, expiry-day sweeps — realized P&L booked to the journal.



Risk That Actually Fires

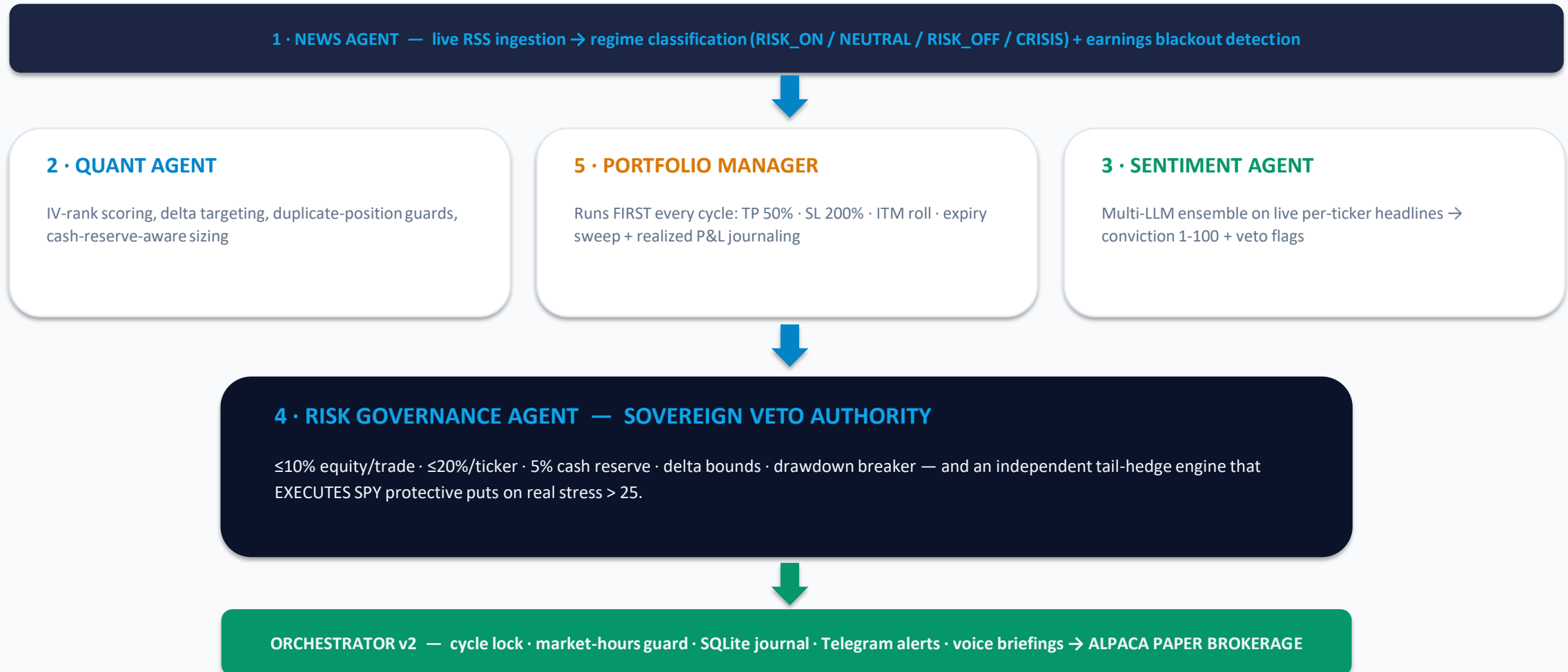
A real SPY ATM-IV VIX-proxy stress index (>25 triggers) drives executed protective-put hedges with a 20h cooldown — plus deterministic veto gates on every trade.



Measured, Not Imagined

SQLite journal records every trade, veto, hedge, and equity snapshot — Sharpe, drawdown, win rate, and profit factor are computed from real history.

Five-Agent Architecture



The Autonomous Cycle — Manage First, Trade Second

1

Market-hours check

Alpaca clock — management-only audits when closed

2

Account audit

Live equity, buying power, Greeks, real stress index

3

MANAGE positions

TP / SL / roll / expiry sweeps, book realized P&L

4

Classify regime

Live RSS headlines → RISK_ON/OFF/CRISIS

5

Screen chains

IV-rank scoring + duplicate guards + cash sizing

6

Sentiment gate

Multi-LLM consensus per ticker + earnings blackout

7

Risk governance

Deterministic veto gates on every proposal

8

Execute honestly

Real Alpaca limit orders; failures = success:false

9

Tail hedge check

Real SPY-IV stress > 25 → buy put (20h cooldown)

10

Journal + alert

SQLite snapshot, Telegram push, voice briefing

Portfolio Manager — The P&L Closer

Rivals open positions; AEGIS closes them. Every open short option is marked to market with live quotes each cycle:

TAKE-PROFIT

50%

Close when mark $\leq$ 50% of entry premium — the classic Wheel profit lock. Realized P&L is journaled instantly.

STOP-LOSS

200%

Exit when the short doubles against us.
Cuts losers before assignment risk compounds.

ROLL

≤ 5 DTE

ITM near expiry? Close and re-strike: the wheel keeps spinning instead of absorbing assignment.

EXPIRY SWEEP

DTE 0

Final-day sweep closes or lets expire worthless — never leaves dust positions on the account.

Every close is one click away for humans, too

The web cockpit shows every open option with entry premium, live mark, P&L %, Greeks, and management state (HOLD / TAKE_PROFIT / STOP_LOSS / ROLL / EXPIRY) — plus a Close Now button that journals the exit.

News Agent — Real-Time Market Intelligence

Live RSS Ingestion (zero API keys)

- ▶ Google News RSS: market-wide + per-ticker headline feeds, stdlib XML parsing
- ▶ 10-minute cache respects rate limits while staying fresh
- ▶ Grounded prompts: every LLM call receives the actual live headlines — no model memory guessing
- ▶ Earnings proximity: regex + LLM detection of imminent quarterly reports

Market Regime Classification

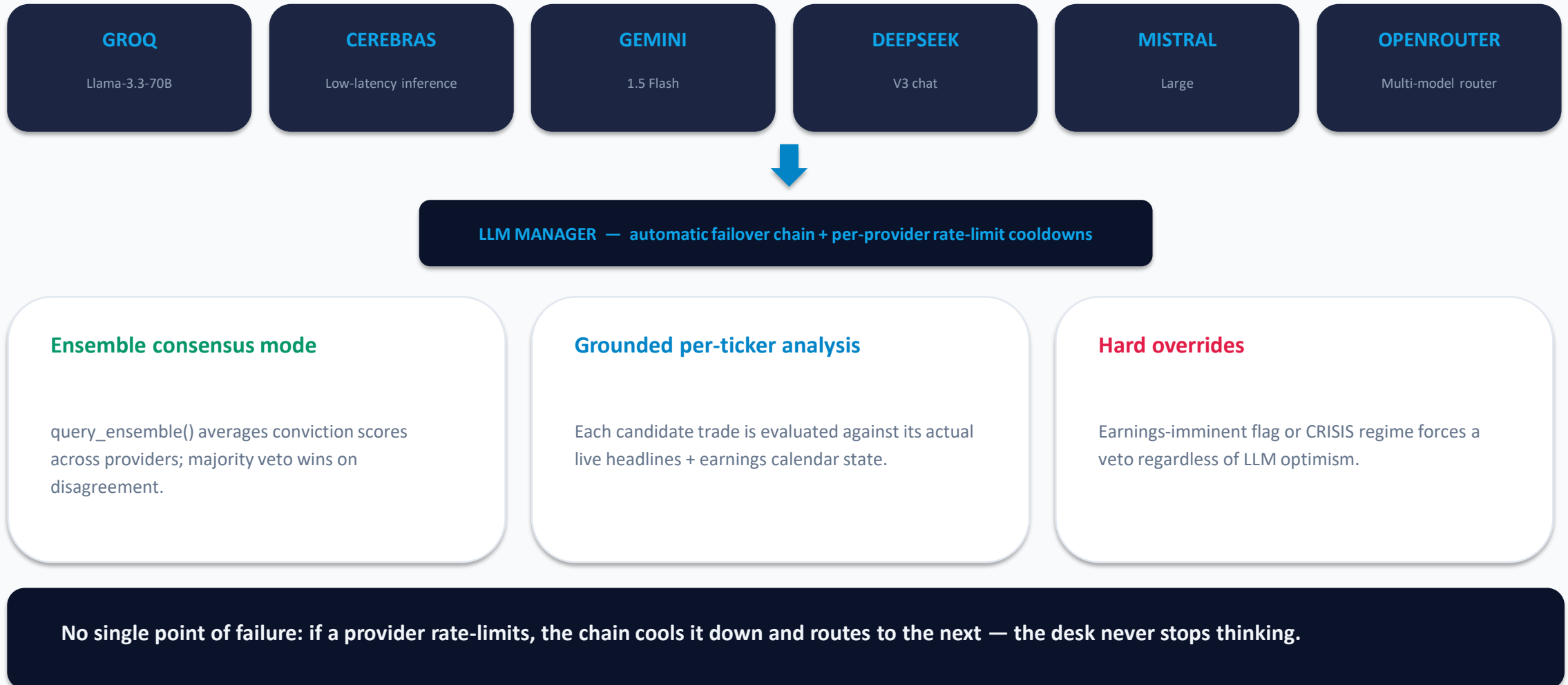
- ▶ RISK_ON / NEUTRAL / RISK_OFF / CRISIS — one LLM call per cycle on live headlines
- ▶ Outputs: confidence, premium-selling favorability, volatility expectation, key risks

LIVE VERIFIED TODAY

"RISK_ON (confidence 85%) — Equities surge despite Fed rate uncertainty... volatility expectation: SUPPRESSED" — computed from live headlines by the ensemble.

Earnings blackout gate: if reports are imminent for a ticker, sentiment is overridden to veto — never sell premium into a binary event.

Sentiment Agent — Multi-LLM Ensemble Consensus



Risk Governance — Sovereign Veto + Real Tail Hedging

🚫 Deterministic Veto Gates

- ▶ Max equity per trade **10%**
- ▶ Max exposure per ticker **20%**
- ▶ Minimum cash reserve **5%**
- ▶ Portfolio delta bounds **[-100Δ, +250Δ]**

🔍 Real Stress Index — not a constant

- ▶ SPY ATM ~30-DTE implied vol → VIX-proxy, computed from live chain data
- ▶ Fallback chain: realized vol from daily bars → last known value
- ▶ Live reading: 12.7 at last cycle — far below the 25 hedge trigger

🔍 The Tail Hedge — actually executes

When stress > 25: find a real ~14 DTE SPY put ~4% OTM from the live chain, buy it at the ask for a reliable fill, and journal the hedge.

A 20-hour cooldown prevents hedge stacking, and every hedge is recorded in the journal with contract, strike, cost, reason, and order id — fully auditable.

SQLite Journal — Every Number Is Measured

trades

open/close lifecycle, premiums, Greeks, verdicts, realized P&L

equity_snapshots

minute-resolution equity curve with stress + regime context

cycle_history

proposals, executions, vetoes, hedges per cycle

hedges

contract, strike, cost, reason, order id

Thread-safe (WAL mode) · zero external DB dependency

Real Analytics Engine

- ▶ Realized P&L (total + per strategy)
- ▶ Win rate, profit factor, avg win/loss
- ▶ Sharpe ratio (from equity-curve returns $\times \sqrt{252}$)
- ▶ Max drawdown from the running peak
- ▶ Equity curve → live dashboard chart

Why judges should care

P&L claims are verifiable against an append-only ledger — every fill, veto, and hedge is timestamped. Restart the process and the desk remembers its book.

Web Cockpit v3.1 — WebSocket Real-Time Cockpit

1

Live Decision Stream

WebSocket pushes every agent event the moment it happens — cycle lock status, vetoes, fills — no 3-second polling.

2

Performance Analytics

Equity curve chart, realized P&L, win rate, Sharpe, drawdown — all rendered from the SQLite journal.

3

Trade Journal

Full blotter of every trade with strategy, verdict, entry/exit, realized P&L.

4

Market Monitor

Live regime pill, stress gauge, watchlist quotes, market clock, RSS news feed.

5

Position Marks

Open options with live marks, P&L %, Greeks, management state + one-click Close Now.

6

Agent Status Bar

Five agent pills pulse with state; market session indicator; auto-loop toggle.

One Desk, Three Doors + Alerts

🔗 Web API v3

- ▶ WebSocket /ws live decision stream
- ▶ 13 REST endpoints performance, journal, news, marks, orders, health
- ▶ POST /run-cycle trigger autonomy on demand
- ▶ POST /toggle-auto 24/7 background loop

🖥️ CLI — 10 Commands

- ▶ --run-cycle · --manage full cycle / manager sweep
- ▶ --performance · --journal real ledger analytics
- ▶ --news · --regime · --status live intel on demand
- ▶ --loop 15 · --json cron-ready autonomy

🔌 FastMCP — 11 Tools

- ▶ Claude / Cursor can run the desk as MCP client
- ▶ Full coverage: account, screen, risk, cycle, hedge, analytics, journal, manage, regime, news, blotter
- ▶ AI-native ops: LLM loops schedule and audit trading

📱 Telegram notifier pushes every execution, veto, hedge, and cycle summary to your phone in real time.

Live Verification — What the System Actually Did

Real cycle transcript (WebSocket stream)

ORCHESTRATOR Market closed → management & audit cycle only

ALPACABROKER Equity \$99,807.80 · Delta 87.26 · 0 \$28/day

PORTFOLIOMGR 3 open positions swept — all healthy, HOLD

NEWSAGENT Regime: RISK_ON (85%) from live headlines

QUANTAGENT 8 candidates · IV-rank scored · dedupe guards

SENTIMENT PLTR 35/100 → VETOED_BY_SENTIMENT

SENTIMENT AMD 75/100 → REJECTED_POSITION_SIZE

SENTIMENT NVDA 85/100 → APPROVED @ \$210 strike

RISKGOV AMZN vetoed · AAPL evaluated

EXECUTION Market closed → deferred to next session

✓ System behaviors verified live

- ▶ Honest execution: order failures return success:false — no fake fills
- ▶ Duplicate guard: never stacked a second put on a held ticker
- ▶ Cycle lock: overlapping trigger → "SKIPPED_OVERLAPPING"
- ▶ Market-closed guard: approved trade deferred, not error-thrown
- ▶ Real stress index: 12.7 computed from live SPY chain IV
- ▶ 16/16 tests: BS pricing, veto gates, hedge trigger, journal math, RSS, routes, MCP tools

Engineering Integrity



Honest Errors

`execute_order` returns `success:false` with the exception — the dashboard, journal, and Telegram all reflect reality. Fake confirmations are worse than failures.



Guards Everywhere

Threading cycle-lock, market-hours gate, duplicate-position map, hedge cooldown, cash-reserve floor — the system protects itself from itself.



Measured Stats Only

Every metric on screen comes from the SQLite ledger. If the desk hasn't done it yet, the screen says zero — not a fantasy backtest number.



Graceful Degradation

No Telegram keys? Skipped. No ElevenLabs? Text briefings. One LLM down? Chain fails over. Every dependency is optional except Alpaca.

How AEGIS Compares

CAPABILITY	TYPICAL BOT	AEGIS v3.1
Position lifecycle	Opens only	Opens → manages → closes (TP/SL/roll/expiry)
News source	Stale LLM memory	Live RSS ingestion, grounded prompts
Market awareness	None	Market clock guard — never trades closed sessions
Risk index	Hardcoded constant	Real SPY ATM-IV VIX-proxy + realized-vol fallback
Tail hedge	Logged only	Executes protective puts, 20h cooldown, journaled
Persistence	Lost on restart	SQLite journal: trades, equity curve, cycles, hedges
Performance stats	Hand-typed	Measured: Sharpe, DD, profit factor, win rate
Duplicate protection	None	Position-state guards block stacking
Delivery	REST polling	WebSocket real-time decision stream
Access	Dashboard only	Web + CLI (10 cmds) + MCP (11 tools) + Telegram

Technology Stack

TRADING & DATA	alpaca-py — option chains, snapshots, real Greeks, market clock, orders
AI INFERENCE	Groq · Cerebras · Gemini · DeepSeek · Mistral · OpenRouter (ensemble + failover)
LIVE DATA	Google News RSS (stdlib XML) · SPY ATM-IV VIX-proxy stress index
PERSISTENCE	SQLite WAL journal — trades, equity curve, cycles, hedges + analytics engine
BACKEND	FastAPI + Uvicorn — 13 REST endpoints + WebSocket hub
PROTOCOL	FastMCP server — 11 tools for AI assistants
ALERTS & VOICE	Telegram Bot API · ElevenLabs TTS briefings
FRONTEND	Vanilla JS cockpit — WS client, charts, journal, monitor (zero build step)
QUALITY	16-test suite · thread-safe design · honest error contract

Roadmap — From Hackathon Desk to Institutional Platform

NOW — v3.1 (Hackathon)

- ▶ Five-agent autonomous Wheel desk on Alpaca paper
- ▶ Full lifecycle management + SQLite journal + live news
- ▶ Voice briefings · command palette · ticker tape · Greeks monitor
- ▶ P&L heatmap · strategy bars · journal search · CSV export
- ▶ Web cockpit · CLI · MCP · Telegram

NEXT — v3.2

- ▶ Historical backtester replaying the same agent pipeline
- ▶ Multi-leg structures (spreads, iron condors) via Risk Agent approval
- ▶ Volatility-surface analytics + per-strategy Kelly sizing

FUTURE — v4.0

- ▶ Live brokerage rails with per-user risk mandates
- ▶ Multi-account orchestration via MCP agent fleets
- ▶ Regenerate-the-deck performance reporting (investor-grade PDF/PPTX)

Design principle on every roadmap item: nothing ships that the Risk Agent cannot veto.

Team ApexArbitrage



Muhammad Zun U Rain Tahir

QUANT & RISK LOGIC

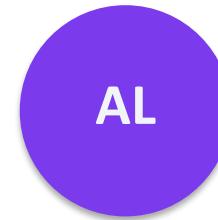
Multi-agent architecture, Black-Scholes Greeks engine, risk governance rules, orchestrator pipeline.



Muhammad Abdullah Khalid

ALPACA INTEGRATION

Alpaca API integration, MCP server, CLI, sentiment ensemble wiring, execution layer.



Anne Loraine A. Pardillo

ANALYTICS & DELIVERY

Web cockpit, performance analytics dashboards, video production, social media presence.

Repository & Deliverables

github.com/MZunurainTahir/aegis · [README](#) · [one-page writeup](#) · [this deck](#) · [video pitch](#) · [social kit](#) · [16-test suite](#)

Judge's Checklist — Why AEGIS Wins



P&L Potential

The Wheel with rule-based exits is the highest-probability premium harvesting design: every open position has a take-profit, a stop-loss, and a roll path — P&L is engineered to be banked, not hoped for.



Creativity & Technical Depth

Five sovereign agents, six-LLM ensemble consensus, live RSS grounding, a real VIX-proxy stress index, executed tail hedges, and a persistent journal — an architecture most retail systems never reach.



Alpaca API Depth

Options chains, snapshot quotes, real Greeks, market clock, order lifecycle, position tracking — with honest error contracts on every call.



Operational Readiness

16/16 passing tests, market-hours guard, cycle lock, duplicate guards, graceful degradation of every optional dependency — it runs unattended for days.

Web Cockpit v3.1 — Voice Intelligence & Command Center

Executive AI Voice Briefing — ElevenLabs Turbo v2.5

One click narrates live equity, net delta, daily theta yield, realized P&L and win rate in a natural voice. Audio is pre-loaded at login and starts instantly from an in-memory cache — regeneration happens in the background, with a volume slider, dedicated regenerate button, status pill, and a Web Speech fallback chain.

Command Palette + Shortcuts

- **Ctrl+K palette:** 13 fuzzy-searchable commands — run cycle, hedge, export, theme, briefing...
- **One-key ops:** R cycle · A auto · H hedge · B briefing · T theme · E export CSV · F fullscreen
- **Esc + ?** closes any overlay / opens the shortcut cheat-sheet modal

Notifications & Awareness

- **Notification center:** unread badge, trade fills, risk vetoes, hedge events — timestamped
- **Sound alerts:** distinct WebAudio tones for trades, vetoes and hedges (toggleable)
- **Dark theme:** one-click institutional night mode, persisted across sessions

Market Telemetry — Ticker Tape, Greeks & Countdown

 Live Ticker Tape — seamless marquee of the 8-name watchlist

SPY · QQQ · AAPL · MSFT · NVDA · AMZN · META · GOOGL scroll with live spot prices and session % changes, green/red colored, hover-to-pause, reduced-motion aware.

 Market Countdown

Live HH:MM:SS countdown to next open/close
parsed from the Alpaca market clock — the desk
always knows the session state.

 Greeks Monitor

Net-delta marker on the $[-100\Delta, +250\Delta]$ band, daily
theta income fill, SPY-IV stress gauge with the 25
hedge-trigger line, and cash-deployed ratio.

 Regime & Session Tiles

RISK_ON / NEUTRAL / RISK_OFF / CRISIS regime pill
and open/closed session indicator refresh with
every cycle.

P&L Attribution & Journal Analytics

1

 Ticker P&L Heatmap

Realized P&L intensity per underlying, computed live from the SQLite journal — instantly shows which tickers pay the desk.

2

 Strategy Performance Bars

Per-strategy realized P&L with trade counts, wins and win rate — signed bars scaled to the largest absolute exposure.

3

 Journal Search + Filters

Full-text search across the blotter, OPEN/CLOSED status filter, live filtered counters and empty-state handling.

4

 One-Click CSV Export

The whole journal exports to a timestamped CSV — quotes, commas and newlines escaped, ready for Excel or the demo video.

5

 Equity Chart Timeframes

All / 50 / 20 point views of the live equity curve let you zoom from the full campaign to the latest sessions.

6

 SVG Icon System

A 40+ symbol inline SVG sprite replaces every emoji — crisp icons at any DPI, vertically centered with labels, zero icon-font dependency.

AEGIS in Action — Live Cockpit Screenshots

Real account state captured from the deployed Railway instance — \$100,283.90 equity, 5 open positions, 1.96 Sharpe, 12.06 stress index

Run AI Cycle

Pause 24/7 Loop (174 cycles)

Test Tail Hedge

Ctrl K

How Aegis Works

AEGIS ALPHA

Autonomous Risk-Governed Options Desk

TR \$182.97 +0.98%

SPY \$771.91 +0.98%

QQQ

EXECUTIVE AI VOICE BRIEFING

Dashboard READY

Portfolio Greeks & Exposure Monitor

RISK ENGINE

NET DELTA +122.1 Δ

THETA / DAY +\$47.86

STRESS INDEX 12.06

CASH DEPLOYED 0.0%

Greeks Monitor

Live Performance — Equity Curve & Analytics (SQLite Journal)

AA 50 20 LIVE JOURNAL DATA

AEGIS Live Equity Curve (% vs session start)

+0.28% Total Return

1.95 Sharpe Ratio

— Profit Factor

0.18% Max Drawdown

Live Market Monitor

NEUTRAL

CLOSED

WATCH: Fed Chair Walsh signals stubborn inflation may require rate hikes in Jackson Hole speech

What to Look Out for in Economic Data This Week (August 31-September 4)

How Kevin Walsh is rewriting the Fed

Fed holds interest rates steady, but 3 officials dissent in favor of a hike

Surging equities and the trickle-up economy

Portfolio Manager — Open Option Marks

5 Options

CONTRACT	QTY	ENTRY	MARK	P&L %	DTE	MANAGEMENT STATE	ACTION
DHTC180909F08852088	-1	\$2.87	\$2.41	+16%	34d	HOLD	—
NVOJ210116F08215888	-1	\$3.95	\$4.99	+1%	41d	HOLD	—
PLTR210116F08215888	-1	\$5.95	\$3.68	+29%	41d	HOLD	—
SOP210116F08215888	-1	\$8.58	\$8.68	+3%	41d	HOLD	—
TSLA210116F08215888	-1	\$9.98	\$8.73	+12%	41d	HOLD	—

Persistent Trade Journal

Export to CSV / Print / All Trades / CSV

TIME	TICKER	STRATEGY	SIDE	PREMIUM	STATUS	REALIZED P&L
14:02:13	NVDA	CASH SECURED PUT	SELL TO OPEN	\$5.85	OPEN	—
13:17:19	TSLA	CASH SECURED PUT	SELL TO OPEN	\$4.85	OPEN	—
13:12:06	PLTR	CASH SECURED PUT	SELL TO OPEN	\$5.85	OPEN	—
13:12:00	SOFT	CASH SECURED PUT	SELL TO OPEN	\$8.58	OPEN	—
13:11:50	INTC	CASH SECURED PUT	SELL TO OPEN	\$2.87	OPEN	—
07:53:18	SPY	TAIL HEDGE	HEDGE BUY	\$6.11	OPEN	—
07:48:43	SPY	TAIL HEDGE	HEDGE BUY	\$6.11	OPEN	—

8 Trades

Position Marks

Performance Analytics

Trade Journal

ApexArbitrage · AEGIS v3.1 | LabLab.ai × Alpaca AI Trading Agents Hackathon

Slide 23 / 24

AEGIS v3.1

The desk that never sleeps, the risk officer that never blinks.

Five agents. Six LLMs. One sovereign risk charter. Zero hardcoded numbers. Every trade proposed, vetoed, hedged, managed, and closed — journaled forever in SQLite.

Discover

IV-rank scoring

Sell

defined-risk premium

Manage

TP · SL · roll · expiry

Hedge

real stress trigger

Journal

measured P&L

Team ApexArbitrage — thank you. Demo: <http://localhost:8000> · Repo: github.com/MZunurainTahir/aegis